

A zebrafish eyeball, moved by six muscles

A Plexus 2 prototype — written to be read, not decoded

2 August 2026

`prototype/eye/`. Plain English throughout.

Organised by what was asked, what was built, and what went wrong on the way — because the things that went wrong are the only part that taught us anything.

1. What was asked, and what it turned into

The request was to simulate an eyeball: four muscles to aim it, a skull to keep it in its socket, two more muscles for torsion, and a multi-panel movie showing deformation, stress and activation. It grew two constraints along the way that changed the model completely.

The eye should deform a little. Not a rigid body on a ball joint — a soft globe with a stiff scleral shell, so muscle tension actually dimples the sclera and the strain runs into the interior. That is what makes MPM the right tool rather than an unnecessary one.

The muscles should be made of particles too, and be seen contracting. This is the constraint that mattered. In the first version a muscle was a line of action and its pull was a body force injected at the insertion patch. Nothing contracted, nothing deformed, and the tendon's grip on the sclera was *asserted*. In the version reported here the muscles are tissue — their own set of material points — and nothing applies a force “to the eye”:

a muscle shortens because an active stress acts along its fibres; it cannot reel itself up because its origin end is anchored to bone; and it moves the eye because its tendon end is embedded in the sclera and the two bodies share one MLS-MPM background grid.

The globe rotates because a muscle got shorter. That is the whole mechanism.

Nothing was promoted. All nine new operators live in `prototype/eye/eye_ops.py` and `prototype/eye/muscle_ops.py`; importing them registers them, and the run is executed by the stock `plexus.schema.load` and `plexus.engine.run`. `src/plexus` is untouched. The deliverable is a `spec.yaml`.

2. The decomposition into sets

set	what it is	state
eye	the globe, as one organ	pos (centroid), gaze = (horizontal, vertical, torsion) in degrees
mpm_particle	the globe's material points (parent eye)	pos/vel + F, C , mass, Lamé, volume
muscle	the six extraocular muscles	act (innervation, the only integrated block), tension , length
muscle_particle	each muscle's material points (parent muscle)	as mpm_particle
orbit	the bony socket	pos , the centre of the cup
mpm_grid (field)	one background grid, shared	—

The shared grid is the point of the design: it is what mechanically couples a contracting muscle to the sclera, and it is free — sharing a background grid is simply how MLS-MPM couples any two bodies. The stock **mpm_scatter** *overwrites* the grid, so a second *implementation* of that same contract, **implementation: accumulate**, was registered: identical biology, different numerics. Per Plexus2 that is the correct way to extend — a new implementation of an existing contract, never an edit to the engine.

The operators

operator	kind	what it does
eye_anatomy	rewire	squash the seeded ball into the ovoid; label sclera / cornea / iris / pupil / choroid / vitreous / lens; per-region Lamé
muscle_morphogenesis	rewire	shape each muscle's points into a tapered strap along its arc of contact; fibre directions; bone cap and tendon cap
eye_pose	aggregate	Kabsch fit of the shell against its rest state → gaze and torsion
muscle_geometry	aggregate	bin points by fibre coordinate → centreline, length , insertion, line of action, rotation axis
oculomotor_drive	lateral	gaze program → PID (the integral term is the oculomotor <i>neural integrator</i>) → rectified projection onto each muscle's axis → activation dynamics
muscle_contract	exchange	$\sigma_a = A a (1 + \beta(\lambda - 1)) \mathbf{f}\mathbf{f}^\top$ along the fibre
bone_anchor	lateral	pin the origin cap to the skull
muscle_sleeve	lateral	the orbital pulley: a purely <i>transverse</i> restoring force, so the path is held but shortening is free
orbit_socket	lateral	penalty contact with the spherical cup + orbital-fat suspension
mpm_scatter [accumulate]	exchange	the second body into the shared grid

The schedule closes a loop: pose readout → gaze error → innervation → contraction → MLS-MPM → motion → pose readout.

3. The one result worth defending

No muscle action is tabulated anywhere in this prototype. Each muscle's rotation axis is re-measured every frame from where its tissue actually is, as $\hat{\mathbf{n}} \times \hat{\mathbf{u}}$ (insertion radius $\times$ line of action), and **oculomotor_drive** simply rectifies the projection of the desired rotation

onto it — Sherrington’s law of reciprocal innervation, one line of code. So the textbook primary/secondary/tertiary actions have to *emerge* from where each muscle inserts and where it pulls from. They do:

muscle	measured axis (x, y, z)	reading	textbook
LR	(0.03, 1.00, -0.05)	pure abduction	abduction only
MR	(0.02, -1.00, 0.03)	pure adduction	adduction only
SR	(-0.90, -0.17, 0.39)	elevation > intorsion > adduction	same order
IR	(0.90, -0.22, -0.37)	depression > extorsion > adduction	same order
SO	(0.61, 0.14, 0.78)	intorsion > depression > abduction	same order
IO	(-0.66, 0.18, -0.73)	extorsion > elevation > abduction	same order

All six agree, in sign and in rank. The medial tilt of the orbital apex is what gives the vertical recti their torsional and horizontal components; a co-axial apex makes SR and IR pure elevators and depressors, which is anatomically wrong. Recruitment measured from the runs agrees with this table on 4 of 4 commands in every mechanically-stable trial.

Why this is worth more than a nice movie. It is a falsifiable prediction that the geometry alone determines the actions, tested against a table nobody consulted while writing the code. It is also the Plexus point in miniature: the mechanism is in the operators and the anatomy, not in a lookup table inside a simulator.

4. Two modelling decisions that are not obvious

The globe is an ovoid; the cup is a sphere. Teleost eyes are flattened along the optic axis. The `mpm_particle` entity can only seed a uniform ball, so the rest configuration is that ball affinely squashed, $z \rightarrow kz$ with $k = 0.82$. An affine image of a uniform ball is a uniform ellipsoid, so density stays uniform, and because this is an *initial condition* and not a deformation, F stays the identity — the ovoid is undeformed and carries no residual stress. The bony cup, by contrast, stays spherical with a radius just above the equatorial semi-axis: an ovoid cannot rotate inside a socket that matches it, and in the animal the gap is filled by orbital fat, not bone.

The fat exerts no torque. The retrobulbar fat pad is modelled as a restoring force toward the socket centre applied *uniformly* to every point of the globe. A uniform body force produces no torque about the centroid, so the fat recentres the eye without ever resisting gaze — which is exactly the division of labour the anatomy has. Without it, the residual normal component of muscle tension slowly walks the globe around inside the cup.

5. The four things that went wrong, and what each one taught

Eleven archived trials, each with its own `spec.yaml`, `movie.mp4` and `diag.json`. The interesting content is the failures.

1. Raising active stress alone (t02) — the muscles ate themselves

At $A/E = 0.8$ the straps shortened by 88% and crushed the globe (strain $\gg 1$). A muscle shortens until its passive tension balances its active stress, so the *steady shortening is set by A/E alone*, while the *force* it delivers is $A \times \text{cross-section}$. Raising A without raising E only buys collapse. **Lesson: A and E are not independent knobs.**

2. Coupling them (t03–t04) — a buckling cliff, and a new ceiling

Held at $A/E = 0.25$ everything is well behaved: 20–27% shortening, 4/4 correct recruitment, 2–5% strain, negligible drift. At $A/E = 0.30$ it collapses again — so the limit is a *buckling* cliff, not a smooth trade-off. But the well-behaved regime capped the eye at 11° . **Lesson: a stiff passive element buys stability by making the antagonist so stiff that it eats most of the agonist’s force.**

3. Geometry (t06–t08) — length is what buys gaze

For a muscle pulling a globe against its antagonist, $\theta_{\max} \sim (A/E) \times (L/R)$. An earlier convenience — truncating each muscle to 55% of its path so the four recti would not project onto one blob — had left the recti at $L/R = 2.3$ and the obliques at 1.15, against ≈ 3.3 in a real orbit. Restoring the length, spreading the rectus origins around the *annulus of Zinn* (a ring, not a point) and widening the sclera stand-off took horizontal range from 12.7° to 29.6° with recruitment still 4/4. **Lesson: the convenience taken for the picture was also a mechanical error.**

4. The force–length relation (t09–t16) — the passive element was backwards

Real muscle is passively *soft* and limits its own shortening through the force–length relation, not through a stiff passive element. Switching on $T = A a (1 + \beta(\lambda - 1))$ means a muscle stops pulling once it has shortened by $1/\beta$, so E can drop far below A and the antagonist gets out of the way: horizontal range jumped to 52° and torsion to 13.7° . But a soft, slender strap then *buckles* — it folds up, losing 60% of its centreline length while the fibre stretch stays near 0.75, so the limiter never engages. The fixes are what the real plant does: a chunkier cross-section (bending stiffness goes as wh^3) and genuine tonic co-contraction, which keeps every muscle in tension and is precisely why extraocular motoneurons fire at high rates even in primary position. **Lesson: two different instabilities, and only one of them was the one being fixed.**

5. The orbital pulley (t15–t22) — the mechanism that was actually missing

A free-floating strap is a column in compression the moment it contracts, and the fix is not numerical: every extraocular muscle passes through a sleeve of collagen and smooth muscle fixed to the orbital wall (Demer’s active-pulley hypothesis), and that sleeve is exactly a *transverse* constraint — it fixes where the path runs without resisting how much the muscle shortens. `muscle_sleeve` applies a restoring force to only the component of displacement perpendicular to the local fibre, tapering to zero over the distal third so the tendon end stays free to follow the rotating globe. Held at everything else constant, it raises the A/E buckling ceiling from ≈ 0.25 to between 0.28 and 0.38 — $A/E = 0.28$ runs clean (20–30% shortening, finite, strain 0.051), $A/E = 0.38$ still folds.

6. The globe came apart (t18) — absolute stiffness, not just contrast

Under the strongest drives the globe stopped being a globe. The vitreous had been set at $E = 9$, a shear modulus of 3.8 — very nearly a fluid. The fixed-corotated law has no failure criterion, so under a hard tendon pull the interior simply flowed and the shell separated from it. Raising the interior to $E = 45$ (vitreous) and 130 (choroid) against a 420 sclera keeps the $\approx 9\times$ core-to-shell contrast the strain panel is there to show, while giving every region enough shear modulus to hold together. **Lesson: “deformable” is a statement about the ratio; “stays in one piece” is a statement about the absolute value.**

The trial ledger

trial	A	E	β	range h/v/t (deg)	short%	strain ₉₉	recruit
t02 viz_fix	48	60	–	22.1 / 12.2 / 4.8	88.6	1.277	2/4
t03 c_a	60	240	–	12.7 / 5.3 / 1.7	27.4	0.052	4/4
t04 c_b	90	300	–	blew up	90.5	1.381	2/4
t06 c_long	55	250	–	26.9 / 11.0 / 3.7	24.3	0.038	4/4
t07 c_longer	55	250	–	29.6 / 11.6 / 4.6	23.9	0.038	4/4
t08 c_wide	55	250	–	29.2 / 11.2 / 4.3	23.8	0.059	3/4
t09 v_fl	130	55	4.0	52.1 / 7.7 / 13.7	61.6	1.067	2/4
t11 v_tone	115	110	4.0	31.9 / 29.3 / 38.9	75.1	1.679	0/4
t15 s_a (sleeve)	90	150	3.0	22.8 / 33.9 / 9.4	56.5	1.684	3/4
t18 q_b	120	190	3.0	globe came apart	54.6	NaN	0/4
t21 r_sleeve70	70	250	–	33.2 / 10.1 / 4.2	30.2	0.051	3/4
t22 r_sleeve95	95	250	–	buckled, NaN	73.6	NaN	1/4

A a peak active stress, E muscle Young’s modulus, β force–length slope. “short%” is peak centreline shortening — above $\sim 38\%$ it is buckling, not contraction. From t21 the `muscle_sleeve` pulley is present and the globe interior is the stiffened one; t21 is the working point the final runs use. Its settled tracking error is 7.3/3.4/2.9° in horizontal / vertical / torsion, against 10–20° for every earlier configuration.

6. Gradient descent on the controller — and why it was refused

`tune_gaze.py` follows `inverse_slime`: differentiating the MPM rollout is exactly the cost Plexus 2 warns about (memory scales as points $\times$ steps), so it never rolls out. It identifies a reduced plant — $\ddot{\theta} = Bu - C\dot{\theta} - K\theta$ per axis, with $u = \sum_m a_m \hat{\mathbf{a}}_m$ read straight from the archived run — by one ridge least-squares fit, then runs Adam on $(k_p, k_i, k_d, \text{gain}, \text{tonic})$ over that differentiable surrogate.

It works mechanically, and the answer is negative, twice over.

First the derivatives were the problem, not the model. The initial fit gave $R^2 = 0.28$ – 0.40 on $\ddot{\theta}$. Differentiating a strided, noisy signal twice amplifies its noise by $\sim h^{-2}$; reading the derivatives off a local Savitzky–Golay polynomial instead raised it to 0.48 – 0.72 with no change to the model.

Then the model was the problem. Tuning still bought $1.008\times$. The first explanation offered — “the drive must be saturated” — was wrong, and the tool’s own diagnostic said so: the agonist sits above 0.98 on only 6% of frames. So the surrogate was asked to reproduce the excursion the real eye actually made, under the gains it actually used:

peak excursion	horizontal	vertical	torsion
real	17.0°	9.0°	2.8°
surrogate	1.7°	3.0°	2.2°

The instrument failed its own check, so the measurement was thrown away. A good R^2 on $\ddot{\theta}$ certifies the *transients*; the standing error we wanted to tune away lives in the steady state $\theta_{ss} = Bu/K$, and the fit misses the real excursion by a factor of ten in the horizontal. Tuning gains on it would be optimising against a model that disagrees with the thing it is a model of. `tune_gaze` now runs this fidelity test and refuses to report a tuned number when it fails — which is the discipline the Plexus2 paper puts first: verify the instrument before trusting the measurement.

The fix is known and not yet done: excite the plant with a frequency sweep rather than a staircase, and fit the steady state explicitly instead of hoping it falls out of an acceleration fit.

7. The movie

Eight panels, black ground, lower-case labels, one shared time cursor.

a	anterior view	the cosmetic eye — white sclera, silvery iris, large black pupil, gold iridophore flecks, after the reference photograph — with the six muscles as their own material points, brightening with activation
b	lateral view	the ovoid profile in the bony cup, the obliques, the trochlea
c	strain	Green–Lagrange $\ E\ $ on a cut through globe <i>and</i> muscles
d	stress	von Mises on the same cut
e	grid momentum	$ v $ on the shared MLS-MPM grid — the medium through which a contracting muscle actually reaches the sclera
f	activation	the six innervations against time
g	length	each muscle as % of its rest length — the contraction, measured
h	gaze	horizontal / vertical / torsion against the command

Panels c–e follow the codebase’s `grid.mp4` diagnostic, extended to two coupled bodies. Two rendering notes that are not cosmetic: the far hemisphere is culled, so the eye reads as a lit solid rather than a point cloud; and muscle points that fall inside the globe’s exact orthographic silhouette *and* behind its centre are hidden, because a depth-sorted scatter alone lets a muscle show through the gaps between globe particles — most visibly straight across the black pupil. **The gold flecks are load-bearing:** a round pupil shows no torsion, so the iridophore specks are what make the roll visible at all.

8. Where it stands, and what is next

Settled. The decomposition into sets; the emergent muscle-action table; the two-body grid coupling via a new *implementation* of an existing contract; the deformable ovoid in a spherical cup with a torque-free fat suspension; the orbital pulley and the A/E ceiling it raises (0.28 clean, 0.32 folds); the globe radius as a metric, an acceptance check and a term in the objective, so “it hit the angle” can no longer be bought by crushing the eye; and the archive discipline — twenty-five trials, each keeping its own spec, movie and numbers.

Open. Reaching the commanded angle rather than about 60% of it. Six rounds each found a real mechanism and a real ceiling, and the configuration that survives all of them (t21) is stable, holds together, contracts realistically and tracks three times better than anything before it — but

it still settles at roughly 60% of a 25° command. The remaining authority has to come from the pulley ceiling, which is known to lie between $A/E = 0.28$ and 0.38 and has not been bisected. The acceptance test is written and automatic: reaches its commands within 6°, torsion above 6°, horizontal above 35°, recruitment 4/4, centroid drift below 6% of the globe radius, globe strain between 0.4% and 12%, and peak shortening between 4% and 38%.

Worth doing next, in order.

1. Finish the working point, then render the long `atlas` program (each antagonist pair driven back and forth three times) and the zebrafish `okr` program at movie resolution.
2. Let the video-captioning VLM watch the result cold and see whether it reports a rolling eye. That is the dynamics-watching critic the Plexus 2 paper names as a required agent, and this is a cheap, honest test of it.
3. Repair the surrogate identification (§6): sweep excitation, explicit steady-state fit. Only then is controller tuning a question worth asking.
4. A real inverse pass: given only the observed gaze trace, recover the six insertion geometries. Loop II on a system whose ground truth we wrote down ourselves.